Rational equations

1

a $x = 0.2$ **b** $x = 0.45, -0.45$ **c** $x = 3.5$ **d** $x = 0.32, -0.32$

e $x = 2.62, 9.38$ **f** $x = -5$ **g** $x = 3.29, -5.17$ **h** $x = 2.53$

2

a $x = -3$

b No it is not, because it makes two of the denominators in the equation equal to 0.

3

a $x = 4$ is not a solution. **b** $x = 4$ is a valid solution.

c $x = 4$ is not a solution. **d** $x = 4$ is not a solution.

e $x = 4$ is an extraneous solution. **f** $x = 4$ is a valid solution.

4

a $x = -\frac{5}{2}$ **b** $x = -5$ **c** $x = 28$ **d** $x = 0$

e $x = 3$ **f** $x = 7$ **g** $x = 5, 2$ **h** $x = -5, -4$

i $x = \frac{3}{2}, \frac{3}{4}$ **j** $x = 2, -13$

5

a $x = 2$ **b** $x = 1$ **c** $x = 4, -4$ **d** $x = 3, -3$

6 $x = 1, -1, -\frac{i}{3}, \frac{i}{3}$

Graphical solutions to rational equations

7

a

i 1 **ii** 1 **iii** 1 **iv** 1

b

i 2 **ii** 1 **iii** 0 **iv** 1

c

i 1 **ii** 1 **iii** 1 **iv** 2

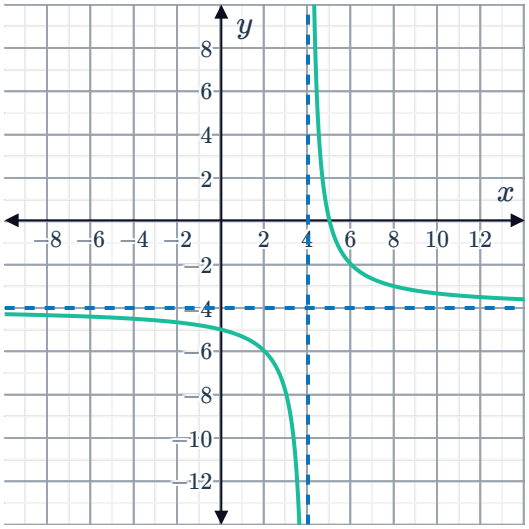
d

i 2 **ii** 1 **iii** 1 **iv** 2

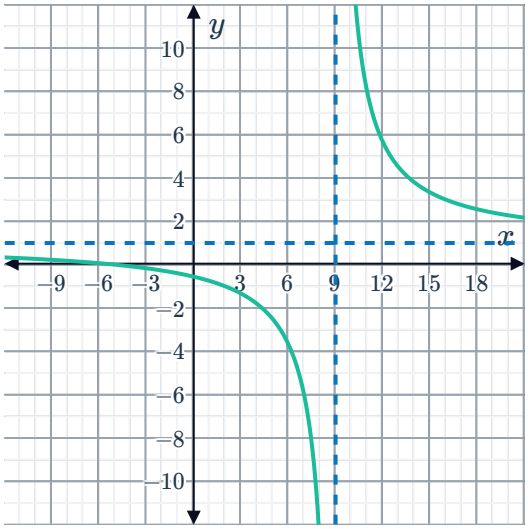
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8

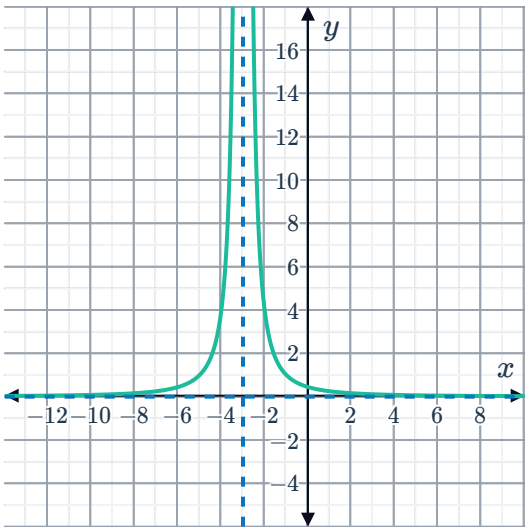
a



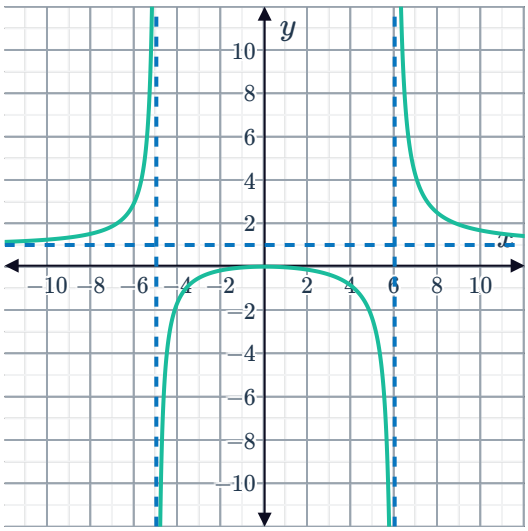
b



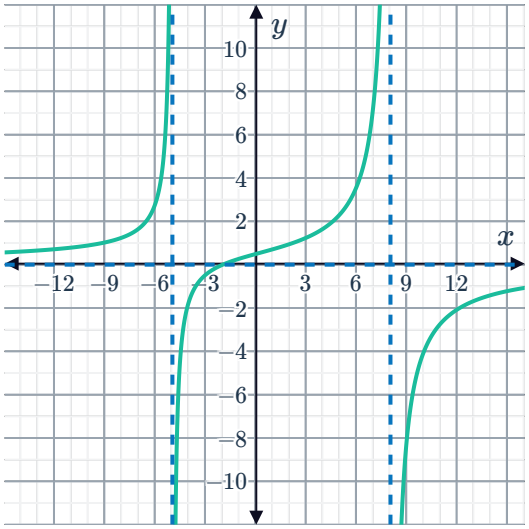
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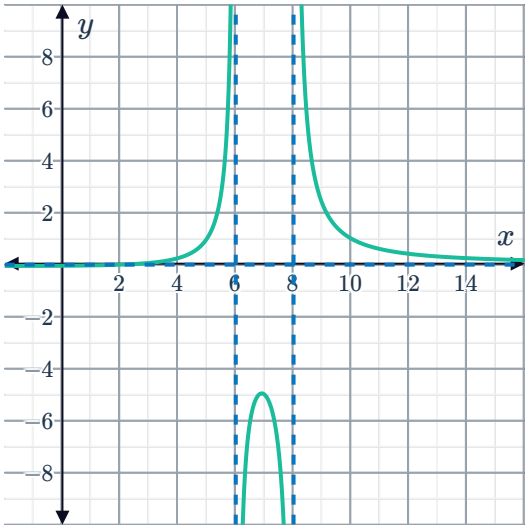
d



e

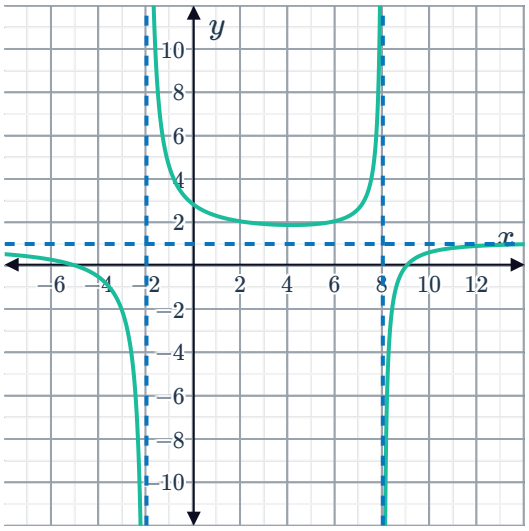


f

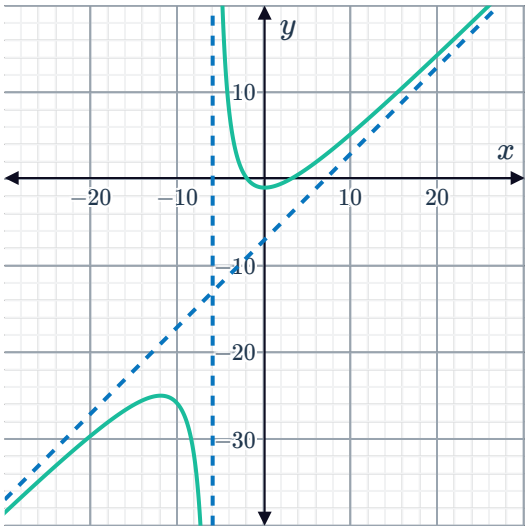


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g

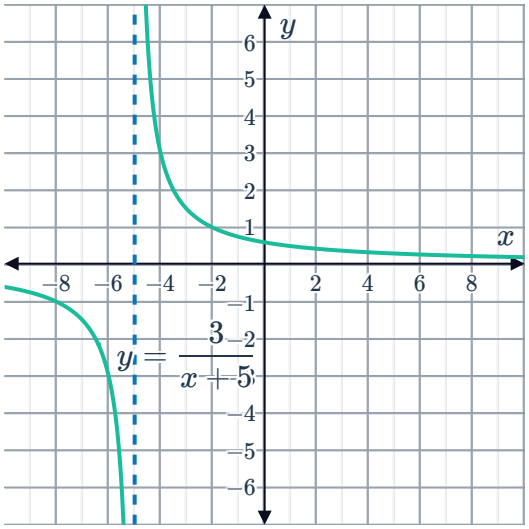


h

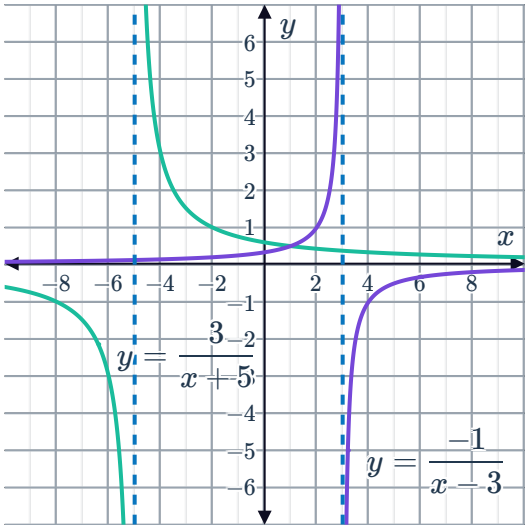


9

a



b

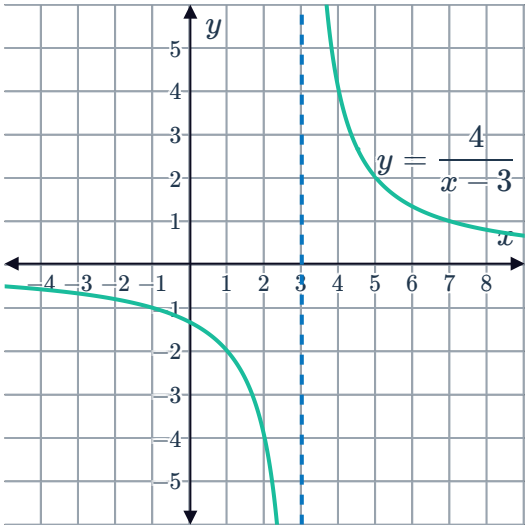


c $x = 1$

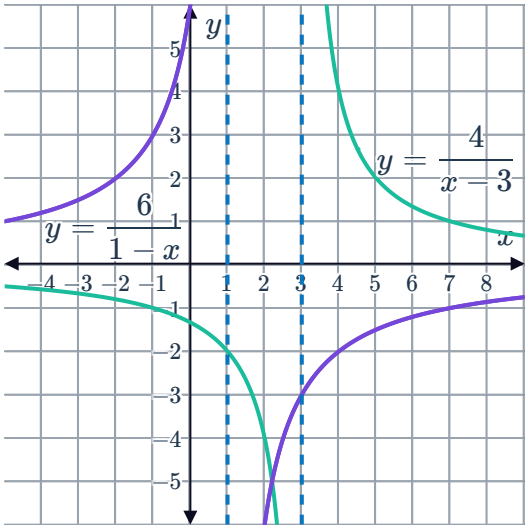
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10

a



b



c $x = 2.2$

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